

```
1 #include <iostream>
2 #include <iomanip>
3 #include <string>
4 #include <fstream>
5
6
7 using namespace std;
8
9 const int SIZE = 31;
10 const int Records = 5;
11
12 struct Students {
13     char name[SIZE];
14     int id;
15     double gpa;
16 };
17
18
19 int main() {
20
21     Students student = { " ", 0, 0.0 };
22
23     fstream data("Data.bin", ios::out | ios::binary);
24
25     cout << "Writing to the file\n";
26
27     for (int i = 0; i < SIZE; i++) {
28         data.write((char*)&student, sizeof(student));
29
30         // data.write(reinterpret_cast<char*>(&student), sizeof(student));
31     }
32     data.close();
33
34     cout << "\nThe size of each student is " << sizeof(student) << endl;
35
36
37     fstream data1("Data.bin", ios::in | ios::binary);
38
39     cout << "\nHere are the data for all the students\n";
40
41
42     data.read((char*)&student, sizeof(student));
43
44
45     while (!data.eof()) {
46         cout << "\nStudent Name: ";
47         cout << student.name << endl;
48     }
```

```
49     cout << "\nID: ";
50     cout << student.id << endl;
51
52     cout << "\nGPA: ";
53     cout << student.gpa << endl;
54
55     cout << "=====\n";
56
57     data.read((char*)&student, sizeof(student));
58
59
60 }
61 cout << endl << endl;
62
63 data1.close();
64
65 int recordNumber{};
66
67
68
69 fstream file("Data.bin", ios::in | ios::out | ios::binary);
70
71 cout << "Which student record would you like to edit";
72
73 cin >> recordNumber;
74 recordNumber += -1;
75
76 file.seekg(recordNumber * sizeof(student), ios::beg);
77
78 file.read((char*)&student, sizeof(student));
79
80 cout << endl;
81
82 cout << "Here is the data for student #" << recordNumber + 1 << "
    before editing:\n";
83
84 cout << "\nStudent Name: ";
85 cout << student.name << endl;
86
87 cout << "\nID: ";
88 cout << student.id << endl;
89
90 cout << "\nGPA: ";
91 cout << student.gpa << endl;
92
93 cout << "=====\n";
94
95 cout << "\nEnter the students # " << recordNumber << " new Data\n";
96
```

```
97
98     cout << "\nStudent Name: ";
99     cin >> student.name;
100
101     cout << "\nID: ";
102     cin >> student.id;
103
104     cout << "\nGPA: ";
105     cin >> student.gpa;
106
107     cout << "=====\n";
108
109
110     file.seekp(recordNumber * sizeof(student), ios::beg);
111
112     file.write((char*)&student, sizeof(student));
113
114     file.close();
115
116     fstream file1("Data.bin", ios::in | ios::out | ios::binary);
117
118     cout << "\nHere are the new data for all the students.\n";
119
120     file1.read((char*)&student, sizeof(student));
121
122
123     while (!file1.eof()) {
124         cout << "\nStudent Name: ";
125         cout << student.name << endl;
126
127         cout << "\nID: ";
128         cout << student.id << endl;
129
130         cout << "\nGPA: ";
131         cout << student.gpa << endl;
132
133         cout << "=====\n";
134
135         file1.read((char*)&student, sizeof(student));
136
137     }
138     cout << endl << endl;
139
140     file1.close();
141
142
143 }
144
```